PRODUCT SPECIFICATION

Sarthi

AI-Powered Government Scheme Assistance Platform

Complete Feature & Sub-Feature Specification

How every feature works, where it appears on the website / app screens, and the resources to build it

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Contents of this document: product snapshot, site map and navigation, 17 feature specifications with sub-features and screen placement, key screen blueprints, user journeys, build priority, and a resource library.

Version 1.0 | 17 October 2026 | Capstone project reference | Prepared in English

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1. Purpose and How to Read This Document

This document lists every feature planned for Sarthi, breaks each one into sub-features, explains how it works step by step, and says exactly where the user will see it (which screen, which part of the screen, on web and on mobile). It is meant to be used by the development team while building, by reviewers during the capstone demo, and as an input for the patent discussion.

Each feature (F1 to F17) follows the same layout so it is easy to scan:

- **Info strip** - where the feature appears, how important it is to build, whether it touches the patent candidate, and who uses it.
- **What it does** - the purpose in two or three sentences.
- **Sub-features table** - each sub-feature, how it works, and where it sits on the screen.
- **Workflow** - the numbered steps from the user's point of view.
- **Backend and data** - the main API calls and tables involved.
- **Edge cases and design notes** - what can go wrong and how the design handles it.

Build priority labels used throughout

Label	Meaning
Core	Must be built and demonstrated. The product does not make sense without it.
Simplified	Build a smaller working version for the demo (for example, 5-10 schemes or a basic UI). Full scale is not needed.
Future	Document it in the report as future scope. Do not spend build time on it now.

Patent relevance labels

Label	Meaning
Patent-core	Part of the one technical module proposed as the patent candidate: OCR document processing, validation, weighted partial confidence score, criterion-level gap identification and localized resolution guidance (Feature F8).
Product feature	A regular product capability. It is deliberately kept outside the patent claim because it relies on common, well-known patterns.
Foundation	Supports the patent module by supplying data or rules, but is not claimed itself.

> **Status note:** The patent module is a candidate concept only. A deeper prior-art search and a review by a registered patent agent are needed before any filing. Similar open-source projects already exist on GitHub (listed in Section 8.7), so the claim must stay tightly focused on the specific combination described in F8.

2. Product Snapshot

India has more than 3,000 central and state welfare schemes, but many eligible citizens never benefit because scheme information is scattered, eligibility language is complex, and nobody guides them through documents and application steps. The official portal myScheme offers profile-based discovery and a three-step flow (enter details, search, select and apply), but it largely stops at showing matching schemes. Sarthi adds a personalised assistance layer on top: explainable eligibility, document-backed scoring, practical local guidance, family-level tracking and proactive alerts.

Design principles

- **Rules decide, AI explains.** Eligibility is always computed by a deterministic rule engine. The AI assistant only retrieves and explains verified scheme content.

- **Preliminary, never final.** Every result is labelled as preliminary screening. The government authority makes the final decision.
- **Official sources first.** Every scheme shows its official link and a last-verified date.
- **Show the gap, not just the verdict.** Instead of a plain Not Eligible, show what is missing and how to fix it.
- **Low-friction for low-literacy users.** Short steps, plain language, Hindi and English, large touch targets, and a low-bandwidth mode.
- **Minimum data, maximum consent.** Collect only what is needed, explain why, and let the user delete it.

Who uses Sarthi

User	What they do in the app	Typical device
Citizen / applicant	Creates a profile, finds schemes, checks documents, follows guidance, applies on the official portal.	Low-cost Android phone, sometimes a shared device
Family manager	Keeps profiles for parents, spouse and children, often from another city, and gets alerts for them.	Smartphone or laptop
CSC / NGO helper (later)	Helps several citizens on a shared device, including offline.	Shared tablet or kiosk
Admin / Editor / Reviewer	Maintains scheme data, rules and sources, and reviews user-reported errors.	Desktop browser

3. Platform Map: Site Map and Navigation

Sarthi is built as one responsive web application that can be installed on a phone as a Progressive Web App (PWA). There is also a separate admin console for staff. The same screens adapt between desktop and mobile.

3.1 Screen inventory

Screen	Route	Who can open it	Purpose
Landing / Home	/	Everyone	Explain the promise, search, browse categories, start the eligibility check.
Sign in / Sign up	/login	Everyone	Mobile OTP login, guest option, language choice.
Consent	/consent	New users	Plain-language data and consent screen shown once after first login.
Profile wizard	/onboarding	Signed-in users	Step-by-step questionnaire that builds the profile.
Profile	/profile	Signed-in users	View and edit profile, completeness meter.
Explore	/explore	Everyone	Search, filters and categories across all schemes.
My Schemes	/my-schemes	Signed-in users	Ranked results with eligibility status and reasons.
Scheme detail	/schemes/:id	Everyone	Benefits, eligibility, documents, steps, sources.
Document check	/schemes/:id/check	Signed-in users	Upload documents, review extracted data, see score, gaps and resolution steps (F8).
Checklist	/checklist	Signed-in users	Document and application checklist across saved schemes.
Apply guide	/schemes/:id/apply	Signed-in users	Step-by-step application guidance and official link.
Assistant	/assistant	Everyone	Full-page AI chat; also a floating chat button on every page.
Saved	/saved	Signed-in users	Bookmarked schemes, notes and comparison.
Family	/family	Signed-in users	Household dashboard and member profiles.
Notifications	/notifications	Signed-in users	Alert centre for new schemes, deadlines and updates.

Screen	Route	Who can open it	Purpose
Settings	/settings	Signed-in users	Language, notifications, privacy, offline data, delete account.
Help	/help	Everyone	FAQs, how it works, contact, report a problem.
Admin console	/admin/*	Staff only	Scheme and rule management, source verification, feedback queue.

3.2 Navigation structure

Surface	Layout
Desktop web	Top bar with logo, language switch, search, notification bell, profile-member switcher and account menu. Left side filters on list pages. Floating chat button at bottom right.
Mobile / PWA	Bottom navigation with five tabs: Home, Explore, My Schemes, Family, Assistant. Top bar with language switch, bell and account. Filters open as a bottom sheet.
Global elements	Offline / online status strip under the top bar, preliminary-result disclaimer on result screens, and a footer with Help, Privacy, Terms and official-source note.
Admin console	Left sidebar: Schemes, Rules, Documents and Offices, Sources, Verification queue, Feedback, Users and Roles, Audit log.

4. Feature Specifications

Features are listed in the order a citizen meets them. F8 is the patent-candidate module and has the most detail.

F1. Onboarding, Authentication and Language

Where it appears	Build priority	Patent relevance	Main users
Landing page, Sign-in screen, language switch in the top bar on every page	Core (simplified login)	Product feature	All citizens, family managers, helpers

What it does. Gets a citizen into the app with as little friction as possible, in their own language, with clear consent about their data.

Sub-features

Sub-feature	How it works	Where on screen
Landing page	Hero section with a one-line promise, a primary button **Check my schemes**, search box, category tiles and a trust strip saying the data comes from official sources with a last-verified date.	First screen at /. Hero at top, categories below, trust strip above the footer.
Language selector	Hindi and English at launch. All text comes from translation files (i18n). The choice is saved in the profile and in the browser.	Top bar on every screen; also a first-run popup.
Mobile OTP login	User enters a 10-digit mobile number, the backend sends an OTP through an SMS gateway, the user enters the 6-digit code, and the app issues a session token with a refresh token.	Sign-in screen /login, or a modal opened from any Save or Apply button.
Guest mode	A quick eligibility check works without an account. The result is kept only in browser memory and the user is asked to sign in to save it.	Link **Continue as guest** under the OTP box.
Consent screen	Explains in plain language what is collected and why. Separate checkboxes for profile data, document processing and notifications. Consent time and version are stored.	Full screen shown once after first login, before the profile wizard.
Session safety	Automatic sign-out after inactivity on shared devices, and a button to clear all data from this device.	Account menu (top right) and Settings > Privacy.

Workflow

1. User opens the landing page and picks a language.
2. User taps **Check my schemes**.
3. User chooses **Continue as guest** or signs in with a mobile OTP.
4. First-time users see the consent screen and accept the items they agree to.
5. User moves to the profile wizard (F2).

Backend and data. POST /auth/otp, POST /auth/login, POST /auth/refresh. Tables: users, sessions, consents (user_id, purpose, version, timestamp).

Edge cases and design notes

- Rate-limit OTP requests per number and per device to stop abuse.
- No Aadhaar number is needed to sign up.
- Keep every sentence at a simple reading level and avoid legal jargon on the consent screen.
- If SMS delivery fails, show a clear retry timer and an alternative such as email OTP.

F2. Profile and Smart Questionnaire

Where it appears	Build priority	Patent relevance	Main users
Profile wizard (full screen), Profile page, completeness card on Dashboard	Core	Foundation	Citizens, family managers

What it does. Collects only the attributes needed to match schemes, one short topic per screen, and skips questions that do not apply.

Sub-features

Sub-feature	How it works	Where on screen
Step-by-step wizard	Five or six short steps: basics, location, income and occupation, category and special status, education, family. One topic per screen, with a progress bar.	Full-screen wizard at /onboarding; progress bar at the top, Back and Next at the bottom.
Dynamic question skipping	The rule set decides which questions are needed. Land size is asked only for farmers; student questions only for relevant ages or occupations.	Invisible to the user; the wizard simply shows fewer steps.
Field helpers	Plain-language tooltips and examples, such as what BPL means or how to estimate annual household income.	Small information icon beside each field.
Prefer not to say	Every sensitive field has a Prefer not to say option. It is stored as unknown and never as a negative answer.	Last option in dropdowns and radio groups.
Completeness meter	Shows the percentage of fields filled and which missing field would unlock more matches, for example Add land size to see 4 more farmer schemes.	Card at the top of Profile and on the Dashboard.
Edit and re-match	Editing the profile re-runs matching automatically. The previous profile snapshot is kept for audit.	Profile page > Edit button.
Voice input (future)	A microphone icon fills text fields from Hindi speech.	Right edge of text inputs.

Workflow

1. User answers each step; progress is saved after every step.
2. The backend validates and normalises the answers (for example, converts income ranges to numbers).
3. The profile is saved and the eligibility engine runs (F4).
4. User lands on My Schemes with the first results.

Backend and data. GET /profile, PUT /profile. Fields: date of birth or age, gender, state, district, rural or urban, annual household income, occupation, education, category (General, OBC, SC, ST), minority status, disability, BPL card, marital status, land holding, family size.

Edge cases and design notes

- Allow resuming an unfinished wizard on another day.
- Validate ranges (age 0-120, income not negative) with friendly messages.
- Never block the user because an optional field is empty; results will show Needs Verification instead.

F3. Scheme Discovery and Search

Where it appears	Build priority	Patent relevance	Main users
Home search bar, Explore tab, scheme list pages	Core	Product feature	Everyone

What it does. Lets any visitor find schemes by keyword, category, life event, state or target group, even without a profile.

Sub-features

Sub-feature	How it works	Where on screen
Keyword search	Full-text search over scheme name, benefits and description with autocomplete and typo tolerance (PostgreSQL full-text search at first).	Search bar at the top of Home and Explore.
Category tiles	Education, Health, Agriculture, Employment, Housing, Women and Child, Senior Citizens, Disability, Business.	Grid on Home; chips on Explore.

Sub-feature	How it works	Where on screen
Life-event browse	Entry points such as Having a baby, Starting a business, Losing a job, Getting married that map to groups of schemes.	Second row on Home.
Filters	State or Central, target group, benefit type (cash, subsidy, loan, insurance), deadline soon, and a Matches my profile toggle.	Left panel on desktop; bottom sheet on mobile.
Scheme cards	Each card shows scheme name, ministry or state, a one-line benefit, a match badge, the last-verified date and a Save icon.	Results list on Explore and My Schemes.
Sorting	Best match, closest deadline, newest.	Dropdown at the top right of the list.

Workflow

1. User types a keyword or taps a category.
2. The backend filters and ranks matching schemes.
3. User refines with filters.
4. User opens a card to see the scheme detail page (F6).

Backend and data. GET /schemes?query=&state;=&category;=■=. Table: schemes with indexed text columns and category tags.

Edge cases and design notes

- The empty state suggests removing filters and offers the assistant.
- Every card carries a small Official source icon linking to the government page.
- Search must work in Hindi and English; store both names for each scheme.

F4. Deterministic Eligibility Engine and Explainable Results

Where it appears	Build priority	Patent relevance	Main users
My Schemes page, Why this result? drawer, scheme detail Eligibility tab	Core	Foundation (feeds the patent module)	Citizens, family managers

What it does. Checks the profile against each scheme's structured rules and explains the outcome criterion by criterion. The AI never makes this decision.

Sub-features

Sub-feature	How it works	Where on screen
Rule evaluation	Every scheme has criteria stored as field, operator, value, mandatory flag and weight. The engine returns PASS, FAIL or UNKNOWN for each criterion.	Runs in the backend whenever the profile is saved.
Overall status	Eligible, Likely eligible (needs verification), or Not eligible. Always labelled Preliminary.	Coloured badge on each scheme card.
Criterion-by-criterion explanation	A checklist such as Age 18-25: PASS (you are 21); Income below Rs. 2.5 lakh: UNKNOWN (not provided).	Why this result? accordion on the card and the Eligibility tab on the detail page.
AND / OR / exclusions	Nested groups such as (Farmer OR Tenant farmer) AND NOT income-tax payer. Built by admins in the rule builder.	Invisible to the user; visible in the admin rule builder.
Missing-information prompts	UNKNOWN criteria create a one-tap prompt such as Add your income to confirm.	Amber banner on the card.
Audit trail	Every evaluation stores the profile snapshot, rule version and timestamp.	Admin audit view; user-facing History in Profile.

Workflow

1. A profile is saved or edited.
2. The engine loads all published schemes and evaluates each rule group.

3 Results are stored with criterion-level outcomes.

4 My Schemes shows the list with badges and reasons; ranking is applied (F5).

Backend and data. POST /eligibility/evaluate. Tables: scheme_rules, eligibility_evaluations (inputs, rule_version, outcomes, timestamp).

Edge cases and design notes

- A failed mandatory criterion can never be shown as Eligible, whatever the other scores are.
- Use UNKNOWN whenever data is missing or automated verification would be unsafe.
- Unit-test every operator: greater-or-equal, less-or-equal, equals, IN, BETWEEN, AND, OR, NOT, missing values.

F5. Recommendation Ranking

Where it appears	Build priority	Patent relevance	Main users
Order of cards on My Schemes and the Top picks card on the Dashboard	Core	Product feature (separate from the patent score)	Citizens, family managers

What it does. Puts the most relevant schemes first using a transparent weighted match, so the user sees the best opportunities without scrolling through everything.

Sub-features

Sub-feature	How it works	Where on screen
Weighted match score	Criteria carry weights (for example occupation 25, income 20, age 15, state 15, education 10, others 15). The score is the weighted share of satisfied criteria.	Shown as a match percentage on each card.
Mandatory gate	Schemes with a failed mandatory rule drop into the Not eligible group regardless of score.	Group headings on My Schemes.
Deadline boost	Schemes closing soon are moved up within their group.	Closing soon tag on the card.
Grouped sections	Eligible now, Almost eligible (one gap away), Explore later.	Section headings on My Schemes.
Reason chips	Small chips such as Matches: Age, State, Occupation explain why a scheme is ranked high.	Under the card title.

Workflow

1 Eligibility results are produced (F4).

2 The ranker computes the weighted score and applies the mandatory gate and deadline boost.

3 The list is grouped and displayed with reason chips.

Backend and data. POST /recommendations. Weights live in the scheme_rules table so editors can tune them.

Edge cases and design notes

- This match score is for ordering only. The document-backed confidence score in F8 is a separate measure.
- Never describe a high match as a guarantee of benefit.

F6. Scheme Detail Page

Where it appears	Build priority	Patent relevance	Main users
Scheme detail screen at /schemes/:id	Core	Product feature	Everyone

What it does. One page that holds everything about a scheme: benefits, eligibility, documents, steps, sources and next actions.

Sub-features

Sub-feature	How it works	Where on screen
Header	Scheme name, ministry or state, status (active or closed), last-verified date, official link, Save and Share buttons.	Top of the page.
Overview and benefits tab	Plain-language summary of what the scheme gives, with the original wording available on request.	First tab.
Eligibility tab	Rule list with the user's PASS, FAIL or UNKNOWN per criterion (F4).	Second tab.
Documents tab	Required documents and the how-to-get-it guide (F7).	Third tab.
How to apply tab	Online and offline steps and the apply guide (F9).	Fourth tab.
Sources and FAQs tab	Official URLs, notification or guideline links and common questions.	Fifth tab.
Primary actions	**Check my documents** (opens F8), **Ask about this scheme** (opens F10 with context), **Apply** (opens F9).	Sticky action bar at the bottom on mobile; right column on desktop.
Report wrong information	Opens a short form that sends the page and the issue to the admin feedback queue (F15).	Footer link on the page.

Workflow

1. User opens a scheme from a card or search result.
2. The page loads scheme data, official sources and the user's cached eligibility result.
3. User reads the tabs and chooses an action.

Backend and data. GET /schemes/{id}. Tables: schemes, scheme_rules, document_requirements, application_steps, sources.

Edge cases and design notes

- Show a banner when a scheme is closed or its data is older than the freshness limit.
- Always show where the information came from and when it was last checked.
- Provide a WhatsApp share link that opens the public scheme page, not private user data.

F7. Document Checklist and Local Document Guide

Where it appears	Build priority	Patent relevance	Main users
Documents tab on the scheme page, My Checklist page	Core	Foundation (supplies resolution data to F8)	Citizens, family managers, helpers

What it does. Turns the list of required documents into a personal checklist and tells the user exactly how and where to get each missing one.

Sub-features

Sub-feature	How it works	Where on screen
Auto checklist	Built from the scheme's document requirements, for example Aadhaar, income certificate, category certificate, bank passbook.	Documents tab and My Checklist page.
Document status	Each item can be marked Have it, Don't have it, or Needs renewal.	Toggle at the right of each item.
How to get it	For each document: issuing office, online portal link, supporting documents to carry, typical processing time and fee, taken from verified data.	Expandable panel under each item.
District-level office directory	Office name, address and timings for the user's district, with a fallback to state-level guidance if the district is unknown.	Inside the How to get it panel, with a map link.
Reminders	A reminder for documents that expire, such as an income certificate valid for a limited period.	Bell notifications and checklist badges.
Export and share	Download the checklist as a PDF or share it with a family member.	Buttons at the top right of the checklist.

Workflow

1. User opens the Documents tab for a scheme.
2. The checklist is created and any previously verified documents are ticked.
3. User marks what they have and opens guidance for the rest.
4. Missing items feed the document check in F8 and reminders in F13.

Backend and data. GET/POST /schemes/{id}/checklist. Tables: document_types, document_requirements, resolution_paths (document type, state, district, office, link, time, fee, last_verified).

Edge cases and design notes

- Resolution data must come from verified official sources and show a last-checked date.
- When district data is missing, show state-level steps and a clear note rather than guessing.

F8. Document Verification and Partial-Eligibility Scoring (Patent Candidate)

Where it appears	Build priority	Patent relevance	Main users
Check my documents flow from the scheme page: Upload screen, Extracted details screen, Eligibility report screen	Core (basic working version, highest polish priority)	**Patent-core**	Citizens, family managers, helpers

What it does. Instead of a plain Yes or No, this module reads the user's own documents, checks them against each scheme criterion, shows a partial confidence score, names the exact gap and tells the user how to fix it locally. This is the technical module proposed as the patent candidate.

Sub-features

Sub-feature	How it works	Where on screen
Upload and capture	Camera capture or file upload (JPG, PNG, PDF) with an on-screen guide for edges and light, size limits and multi-page support. Files are scanned and stored encrypted.	Upload screen with a document-type picker and a Take photo button.
Image pre-processing	Deskew, denoise, contrast and resolution normalisation before OCR.	Backend only; the user sees a short Reading your document progress state.
OCR extraction	Tesseract (Hindi and English) or a cloud OCR API reads the image. Per-document-type extraction rules pull out name, date of birth, address, income figure, category, issuing authority, issue date and certificate number.	Extracted details panel next to the document preview.
User confirmation	Low-confidence fields are highlighted. The user can correct any field; corrections are stored.	Editable fields with a confidence icon beside each.
Field validation	Checks expiry or currency, name match across documents (fuzzy), numeric plausibility and format patterns. Only the last four digits of an Aadhaar number are kept.	Validation list with severity icons under the extracted details.
Criterion-level comparison	Validated fields are mapped to the scheme's criteria. Each criterion becomes Satisfied, Unmet or Uncertain.	Criteria table in the Eligibility report.
Weighted partial confidence score	Score = sum of weights of satisfied criteria divided by the sum of all weights, times 100. A hard-disqualifier is reported separately so a high score never implies eligibility.	Gauge at the top of the report labelled Preliminary, for example 75 percent.
Gap identification	Lists the exact unmet or uncertain criteria and the reason for each.	What's missing cards under the gauge.
Localized resolution path	For each gap, an action card says what to obtain, where (office or portal for the user's district), what to carry, typical time and a link. Data comes from F7.	How to fix this panel inside each gap card.
Re-score on new upload	When the user uploads a corrected or missing document, only the affected criteria are recomputed and the score updates, for example 75 percent to 100 percent.	Upload missing document button on a gap card.

Sub-feature	How it works	Where on screen
Report export	A PDF summary of score, gaps and next steps that can be shared with family or a helper.	Download icon at the top right of the report.

Workflow

1. User opens a scheme and taps **Check my documents**.
2. User picks a document type and uploads or photographs it.
3. The backend pre-processes the image and runs OCR.
4. The user reviews and corrects the extracted fields.
5. Validation runs and the fields are compared with each criterion.
6. The weighted score is calculated and gaps are identified.
7. Each gap is mapped to a localized resolution action for the user's district.
8. The Eligibility report shows the score, the gaps and the fix steps.
9. The user uploads the missing document and the score is recomputed.

Backend and data. POST /documents, POST /documents/{id}/verify, POST /eligibility/{scheme_id}/document-evaluation. Tables: uploaded_documents, extracted_fields (value, confidence, corrected_by_user), eligibility_evaluations (score, per-criterion status), resolution_paths.

Edge cases and design notes

- Use sample or dummy documents for the demo. Never use real people's documents in public demos.
- If OCR fails or confidence is low, fall back to manual entry so the user is never stuck.
- Never present the score as an approval. Always show the Preliminary label and the official link.
- Set a retention rule: delete uploaded images after processing or after a fixed number of days, and keep extracted fields only if the user chooses to save them.
- For the patent discussion, describe the pipeline as one connected technical process (evidence extraction, validation, weighted comparison, gap detection, localized resolution). Share the similar GitHub projects with the patent agent.
- Do not publish this module in a public repository or demo video before the patent filing decision is taken.

F9. Guided Application Assistance

Where it appears	Build priority	Patent relevance	Main users
How to apply tab, Apply guide screen at /schemes/:id/apply, Application tracker on the Dashboard	Simplified	Product feature	Citizens, family managers, helpers

What it does. Takes the user from I am eligible to I have applied, without leaving them alone on the official portal.

Sub-features

Sub-feature	How it works	Where on screen
Readiness check	Before applying, shows which mandatory documents are ready and which are missing, with links to F7 and F8.	First card on the Apply guide.
Step-by-step guide	Mirrors the official application form section by section and shows what to enter in each field, using profile data with copy buttons.	Stepper at the centre of the Apply guide screen.
Official portal handoff	Opens the official application page in a new tab with a reminder that Sarthi does not submit on the user's behalf.	Primary button at the bottom of the guide.
Offline route	Find the nearest CSC or government office with address and timings for users who prefer to apply in person.	Secondary card with a map link.

Sub-feature	How it works	Where on screen
Application tracker	The user records status (Not started, Applied, In review, Approved, Rejected), application number and date. Self-reported unless an official status API exists.	Tracker card on the Dashboard and the scheme page.
Assisted apply (future)	A trained helper opens the guide on behalf of a citizen with the citizen's consent.	Helper mode in the account menu.

Workflow

1. User taps Apply on a scheme.
2. The readiness check confirms documents.
3. The user follows the step-by-step guide.
4. The user opens the official portal and submits there.
5. The user records the application number and status in the tracker.

Backend and data. Tables: application_steps, applications (user_id, scheme_id, status, reference_no, applied_on). No portal passwords are ever stored.

Edge cases and design notes

- Do not claim to submit applications. Sarthi guides; the official portal receives the application.
- Remind users to keep their application number safe and offer to save it.

F10. AI Assistant (RAG Chatbot)

Where it appears	Build priority	Patent relevance	Main users
Floating chat button on every page, Assistant tab, Ask about this scheme on the detail page	Simplified (5-10 schemes first)	Product feature	Everyone

What it does. Answers natural-language questions in Hindi, English or Hinglish, using only verified scheme content, and always shows its sources.

Sub-features

Sub-feature	How it works	Where on screen
Natural-language Q&A;	Understands questions such as I am a 21-year-old student with family income of Rs. 2 lakh, what can I explore?	Chat window; opens from the floating button or the Assistant tab.
Grounded retrieval	Scheme text is split into chunks and embedded in a vector store (FAISS or Chroma). Structured filters such as state and category narrow the search before semantic retrieval.	Backend only.
Source citations	Every answer ends with chips showing scheme name, section, official URL and last-verified date.	Under each answer bubble.
Safe refusal	If the retrieved evidence is weak, the assistant says it cannot confirm and points to the official source instead of guessing.	In the answer text.
Eligibility questions	For eligibility the assistant calls the rule engine result (F4) and explains it. It does not decide eligibility itself.	Answer bubble with a Check my eligibility button.
Suggested questions	Context-aware chips such as What documents do I need? and When is the deadline?	Above the input box.
Feedback	Thumbs up or down on each answer; down-votes go to the admin queue.	Right of each answer.
Offline state	When the device is offline the assistant shows Assistant needs internet rather than failing silently.	Chat window banner.

Workflow

1. User types or speaks a question.

2 The backend detects the language and applies structured filters.
3 Top matching chunks are retrieved from the vector store.
4 The language model writes an answer using only that context.
5 A citation check confirms the sources and the answer is shown with chips.

Backend and data. POST /assistant/query. Components: embeddings, FAISS or Chroma index, LLM API, prompt template with a do-not-invent rule, query log table (no personal data).

Edge cases and design notes

- Build a test set of 30-50 questions with known answers and check grounding before the demo.
- Log unanswered questions so editors can add missing scheme content.
- The prompt must forbid inventing criteria, benefits, documents or links.

F11. Saved Schemes, Notes and Comparison

Where it appears	Build priority	Patent relevance	Main users
Save icon on every card, Saved tab at /saved	Simplified	Product feature	Citizens, family managers

What it does. Lets users keep a short list of schemes to come back to, add notes, and compare two schemes side by side.

Sub-features

Sub-feature	How it works	Where on screen
Save and unsave	Tap the heart on any card or detail page. Saved schemes sync across devices for signed-in users.	Heart icon on cards and the detail header.
Notes and tags	A private note per saved scheme and an optional tag for which family member it is for.	Saved tab, below each saved card.
Compare	Pick two schemes to view benefits, eligibility and documents in two columns.	Compare button on the Saved tab.
Share	Share a public scheme link through WhatsApp or SMS.	Share icon on the card.

Workflow

1 User taps Save on a scheme.
2 The scheme appears in the Saved tab with a note field.
3 User optionally selects two saved schemes and opens Compare.

Backend and data. GET/POST/DELETE /saved-schemes. Table: saved_schemes (user_id, scheme_id, member_id, note).

Edge cases and design notes

- Saved schemes also feed notifications: changes to a saved scheme trigger alerts (F13).

F12. Family Dashboard

Where it appears	Build priority	Patent relevance	Main users
Family tab at /family, member switcher in the top bar	Simplified (UI and data model)	Product feature (not claimed)	Family managers, citizens

What it does. One account manages profiles for parents, spouse and children, so the whole household can be checked and tracked in one place.

Sub-features

Sub-feature	How it works	Where on screen
Add a family member	Name, relationship, date of birth and the key fields needed for matching. Shared household fields such as address, state and household income are inherited and can be edited.	Add member button on the Family tab; opens a short form.
Household overview	A card per member showing the number of eligible schemes, almost-eligible schemes and pending documents.	Grid on the Family tab.
Member switcher	Switch the whole app to a member's context to see their schemes, checklist and notifications.	Avatar dropdown in the top bar.
Per-member results	Same results, checklist and document check flows as the account owner, but for the selected member.	My Schemes, Checklist and Document check screens.
Household top opportunities	A combined list of the best schemes across all members.	Top card on the Family tab.
Remote care mode	A child living in another city manages a parent's profile and receives the parent's alerts.	Same screens; notification text names the member.
Permission note	A required checkbox confirming the owner has this person's permission to store their details.	Add member form.

Workflow

1. Owner opens the Family tab and adds members.
2. The engine runs for each member (F4).
3. The overview shows counts per member.
4. Owner switches to a member to review schemes and documents.

Backend and data. Table: family_members (owner_id, relation, profile fields, consent flag). Evaluations are stored per member.

Edge cases and design notes

- Collect the minimum data per member and allow deletion of any member at any time.
- Children's data needs extra care. The DPDP Act has a specific section on processing children's data, so get legal advice before storing data of anyone under 18.
- Later versions can let a family co-manager join using their own OTP.

F13. Proactive Notifications

Where it appears	Build priority	Patent relevance	Main users
Bell icon in the top bar, Notifications page, Settings > Notifications	Simplified (in-app first)	Product feature (not claimed)	Citizens, family managers

What it does. Watches for new and changing schemes and tells the user in time, so no eligible family member misses an opportunity or a deadline.

Sub-features

Sub-feature	How it works	Where on screen
Scheme watcher	A scheduled job (for example daily) detects new, updated or closing schemes and re-runs the eligibility engine against every saved profile and family member.	Backend scheduler; no screen.
Trigger types	New scheme, changed criteria, deadline in 15, 7 and 1 days, saved scheme updated, a document about to expire.	Notification cards, grouped by type.
Notification card	Example: New scheme X: Mom may be eligible. Review now. Each card deep-links to the scheme or the result.	Notifications page and bell dropdown.
Channels	In-app bell first. SMS and WhatsApp later. Email optional.	Settings > Notifications.

Sub-feature	How it works	Where on screen
Preferences	Choose channels, quiet hours, language and which family members to follow.	Settings > Notifications.
Digest or instant	Users can choose a daily digest instead of instant alerts to avoid overload.	Settings > Notifications.
De-duplication and limits	The same scheme is never sent twice for the same member, and daily send limits apply.	Backend only.

Workflow

1. A scheme is added or changed, or a deadline gets close.
2. The watcher finds profiles that may match using the rule engine.
3. A notification is created for each affected member and sent on the chosen channels.
4. The user taps it and lands on the scheme or result screen.

Backend and data. Scheduler (cron, APScheduler or Celery beat). Table: notifications (user_id, member_id, type, scheme_id, status, sent_at). GET/POST /notifications.

Edge cases and design notes

- Wording must say may be eligible and never promise a benefit.
- Respect consent: only send on channels the user agreed to, and always allow opt-out.
- Start with in-app notifications only. Real SMS and WhatsApp delivery need paid providers and approvals.

F14. Offline-First Mode

Where it appears	Build priority	Patent relevance	Main users
Install prompt, Offline status strip under the top bar, Settings > Offline data	Future (demo a small subset only)	Product feature (not claimed)	Rural users, helpers on shared devices

What it does. Lets users complete the questionnaire and get a preliminary eligibility result without internet, then syncs when the connection returns.

Sub-features

Sub-feature	How it works	Where on screen
Installable PWA	The app can be added to the home screen and opens even without internet.	Install prompt banner; Add to Home Screen.
Offline app shell	A service worker caches the app's screens and static files.	Invisible to the user.
Local rule cache	A versioned pack of scheme rules for the chosen state is stored in the browser database (IndexedDB).	Settings > Offline data > Download pack by state.
Offline evaluation	The same PASS, FAIL, UNKNOWN logic runs on the device so results match the server.	My Schemes works as usual with an offline strip.
Status strip	Shows Offline - using data from a date and warns when the cached pack is old.	Strip under the top bar.
Sync on reconnect	Pending profiles and results are uploaded, the rule pack is refreshed, and the server re-validates the result.	A Syncing indicator, then a confirmation toast.
Disabled when offline	AI assistant, document OCR and notifications show a clear Needs internet message.	Greyed buttons with an explanation.

Workflow

1. User downloads the rule pack while online.
2. Later, with no signal, the user completes the questionnaire.
3. The local engine returns a preliminary result.

4 When online again, the app syncs and shows any changes from the server.

Backend and data. GET /rules/sync?since_version=, POST /offline/reconcile. Client storage: IndexedDB for rules and pending results; encrypted cached profile.

Edge cases and design notes

- Keep one set of rules as JSON and evaluate it with identical logic in Python and JavaScript, and test that both give the same results.
- Clear cached personal data on logout and on shared devices.
- Treat this as an engineering feature; it is deliberately not part of the patent claim.

F15. Admin Console

Where it appears	Build priority	Patent relevance	Main users
Separate staff site at /admin with its own login	Core (minimal version)	Foundation	Admin, Editor, Reviewer

What it does. Lets staff keep scheme data accurate, structured and fresh, which is the backbone of the whole product.

Sub-features

Sub-feature	How it works	Where on screen
Scheme CRUD	Create, edit, publish and retire schemes with all fields: names in two languages, benefits, target group, scope, status, official URL.	Schemes list and edit form.
Rule builder	Add criteria rows (field, operator, value, mandatory, weight) and group them with AND and OR. Preview the effect on sample profiles.	Rules tab inside a scheme.
Documents and offices editor	Maintain required documents and the district-level resolution paths used by F7 and F8.	Documents and Offices screens.
Sources and provenance	Record the official URL, source type, who verified it and when.	Sources tab inside a scheme.
Verification workflow	Draft, In review, Published. A second person can be required to approve before publishing.	Status dropdown and Verification queue.
Freshness dashboard	Lists schemes not verified within a set number of days and broken official links.	Dashboard home.
Feedback queue	Reports from users (wrong information, bad answers) with status and assignee.	Feedback screen.
Versioning and rollback	Every rule change creates a new version; evaluations record which version they used. Roll back if needed.	History tab.
Roles and audit log	Role-based access (Admin, Editor, Reviewer) and a log of every change.	Users and Roles, Audit log.

Workflow

1 Editor adds or updates a scheme and its rules from an official source.
2 Reviewer checks the source and approves.
3 The scheme is published and appears in search and matching.
4 Freshness alerts remind the team to re-verify later.

Backend and data. CRUD /admin/schemes, /admin/rules, /admin/sources. Tables: scheme_versions, rule_versions, audit_log, users with roles.

Edge cases and design notes

- No silent edits to production data; every change is logged and attributable.

- For the capstone, a minimal admin (scheme form plus rule rows) is enough. Seed data can also be loaded from a file.
- Verify 30-50 schemes by hand from official sources. Do not rely on AI-generated scheme data without checking.

F16. Trust, Privacy and Safety

Where it appears	Build priority	Patent relevance	Main users
Settings > Privacy, banners on result screens, footer links, consent management page	Core (basics)	Product feature	Everyone

What it does. Makes the product trustworthy for a sensitive domain by being clear about data, sources and limits.

Sub-features

Sub-feature	How it works	Where on screen
Source and freshness labels	Official source link and last-verified date on every scheme and answer.	Scheme header, cards and assistant answers.
Preliminary disclaimer	A short banner that results are preliminary and the government decides.	Top of My Schemes, Eligibility tab and Eligibility report.
Consent management	View, change or withdraw each consent at any time.	Settings > Privacy.
Data export and delete	Download my data and delete my account and documents.	Settings > Privacy.
Document retention	Uploaded images are deleted after processing or a fixed period; sensitive numbers are masked.	Explained in Settings > Privacy; automatic in the backend.
Accessibility	Large-text mode, high contrast, screen-reader labels and a low-bandwidth mode that skips heavy images.	Settings > Accessibility; Accessibility icon in the top bar.
Security basics	HTTPS, hashed admin passwords, role-based access, input validation, rate limiting and no sensitive data in logs.	Backend and infrastructure.

Workflow

1. The user sees clear labels and disclaimers throughout the journey.
2. The user can open Settings > Privacy to review consents, export data or delete the account.
3. The backend applies retention and masking rules automatically.

Backend and data. Tables: consents, data_requests, audit_log. Scheduled job for retention clean-up.

Edge cases and design notes

- India's Digital Personal Data Protection Act and the 2025 Rules apply to personal data processing. Get a legal review of consent wording and retention periods before any real launch.
- Do not sell personal data and do not use it for advertising.

F17. Feedback, Community Proof and Help

Where it appears	Build priority	Patent relevance	Main users
Footer of the scheme page, Help tab at /help, Assistant answer thumbs	Simplified (feedback); Future (success stories)	Product feature	Everyone

What it does. Builds trust and improves data quality by letting users confirm, correct and learn from others.

Sub-features

Sub-feature	How it works	Where on screen
Was this correct?	Thumbs up or down on every scheme page and assistant answer. Down-votes ask for a short reason.	Bottom of the scheme page; right of assistant answers.

Sub-feature	How it works	Where on screen
Report an error	A simple form that sends the page and issue to the admin queue (F15).	Footer link on every scheme page.
Help centre	How it works, FAQs, language help and contact details.	Help tab and footer.
Verified success stories (future)	With explicit consent, short stories of people who received a benefit.	Home page strip and scheme page card.
Nearest helper	Link to find a CSC or helper near the user.	Help tab and Apply guide.

Workflow

1. User gives feedback or reports a problem.
2. It lands in the admin feedback queue.
3. An editor checks the source, fixes the data if needed and closes the report.

Backend and data. POST /feedback. Table: feedback (type, scheme_id, message, status, assignee).

Edge cases and design notes

- Never publish a success story without written consent and removal of personal details.

5. Key Screen Blueprints

These blueprints describe what is in each region of the main screens. They can be used directly as a brief for wireframes in a design tool.

5.1 Home (landing)

Region	Content
Top bar	Logo, language switch, search icon, Sign in or account menu.
Hero	Headline promise, one-line subtext, primary button Check my schemes, secondary link Browse all schemes.
Search	Large search box with autocomplete.
Categories and life events	Two rows of tiles.
How it works	Three short steps: tell us about you, see your schemes, get guided to apply.
Trust strip	Official sources, last verified date, privacy promise.
Footer	Help, Privacy, Terms, Contact.

On mobile: *Hero and search stack vertically, categories become a two-column grid, and the bottom navigation appears.*

5.2 Profile wizard step

Region	Content
Top	Progress bar and step title, for example Step 3 of 6: Income and work.
Body	One to three questions with large inputs, tooltips and a Prefer not to say option.
Bottom	Back and Next buttons; Save and finish later link.

On mobile: *Full-width inputs and large buttons pinned to the bottom of the screen.*

5.3 My Schemes (results)

Region	Content
Header	Member switcher, completeness card (for example Profile 70 percent complete, add land size to see more), disclaimer banner.
Section: Eligible now	Scheme cards with badge, benefit line, reason chips, Why this result? link, Save icon.
Section: Almost eligible	Cards that show the single missing item and a Check my documents button.
Section: Explore later	Collapsed list of other schemes.
Filters	Left panel on desktop with category, state or central, deadline.

On mobile: *Filters open as a bottom sheet; cards use the full width.*

5.4 Scheme detail

Region	Content
Header	Name, ministry, status badge, last verified date, Official source link, Save and Share.
Tabs	Overview, Eligibility, Documents, How to apply, Sources and FAQs.
Right column (desktop)	Your result card with status and match percentage, and action buttons: Check my documents, Ask about this scheme, Apply.
Footer of page	Was this correct? and Report wrong information.

On mobile: *The right column becomes a sticky bottom action bar.*

5.5 Eligibility report (document check result)

Region	Content
Top	Scheme name and the Preliminary label, with a download icon.

Region	Content
Score gauge	Large percentage gauge, for example 75 percent, with a one-line message such as You are one document away.
Criteria table	Each criterion with weight, status (Satisfied, Unmet, Uncertain) and the evidence used, such as Income Rs. 1.8 lakh from your income certificate.
What's missing cards	One card per gap with a plain explanation.
How to fix this	Inside each gap card: where to go, documents to carry, typical time, link, and an Upload missing document button.
Footer actions	Re-check, Save report, Go to Apply guide.

On mobile: *Gauge on top, criteria as stacked cards instead of a table, and a sticky Upload button.*

5.6 Family dashboard

Region	Content
Top	Household summary and the Add member button.
Member cards	Avatar, name, relation, count of eligible and almost-eligible schemes, pending documents, and View schemes button.
Household top opportunities	Combined list across members with the member name on each card.
Alerts	Latest notifications for the household.

On mobile: *Member cards stack as a vertical list; the member switcher sits in the top bar.*

5.7 Notification centre

Region	Content
Filters	All, New schemes, Deadlines, Updates, by member.
Notification list	Cards with icon, headline, member name, time and a Review button.
Footer link	Notification settings.

On mobile: *A single-column list; the bell in the top bar shows the unread count.*

6. End-to-End User Journeys

Journey A: A farmer on a low-cost phone

1. Opens the site in Hindi, taps Check my schemes and continues as guest.
2. Answers four short questions: age, state, occupation (farmer), land size.
3. Sees three eligible schemes and two almost-eligible schemes with reasons.
4. Opens one scheme, sees that the income certificate is missing and reads where to get it in his district.
5. Signs in with an OTP to save the result and gets a reminder when the certificate needs renewing.

Journey B: A student checking documents

1. Signs in, completes the profile and opens a scholarship scheme.
2. Taps Check my documents and photographs the Aadhaar, income certificate and category certificate.
3. Corrects one low-confidence field and sees the report: 75 percent, with the category certificate missing.
4. Follows the fix steps, uploads the certificate and sees 100 percent.
5. Opens the Apply guide, applies on the official portal and records the application number.

Journey C: A son in another city managing his parents

1. Adds his father and mother in the Family tab and confirms he has their permission.
2. Sees that his mother may be eligible for a pension scheme and his father for a health scheme.
3. Receives a notification a week later: a new scheme opens and his mother may be eligible.

4 Uses the checklist to arrange documents and asks the assistant a question about the deadline.

7. Build Priority and Phasing

The full list above is a complete product vision. For the capstone timeline, build the Core items fully, build Simplified items as working demos, and document Future items in the report. AI coding tools speed up coding a lot, but data verification, OCR tuning, integration testing and the patent process still take real time.

Feature	Priority	Effort	Why / how to scope it
F1 Onboarding and login	Core	Low	Use mobile OTP with a sandbox SMS provider or a simple email login for the demo.
F2 Profile wizard	Core	Medium	Six steps with dynamic skipping for a few fields.
F3 Discovery and search	Core	Low-Medium	PostgreSQL full-text search is enough.
F4 Eligibility engine	Core	Medium	The heart of the product. Test every operator.
F5 Ranking	Core	Low	Weighted score plus mandatory gate.
F6 Scheme detail	Core	Medium	Five tabs fed from the scheme tables.
F7 Checklist and local guide	Core	Medium	Seed office and process data for one state and a few districts.
F8 Document verification and scoring	Core (basic)	High	The differentiator. Use sample documents, Tesseract, and three to four document types.
F15 Admin console	Core (minimal)	Medium	A scheme form and rule rows; seed data from a file if short on time.
F16 Trust and privacy basics	Core (basics)	Low-Medium	Disclaimers, consent, delete account, masking.
F9 Guided application	Simplified	Medium	Readiness check, steps for two schemes, official link, self-reported tracker.
F10 AI assistant	Simplified	High	Only 5-10 schemes in the vector store, with a 30-question test set.
F11 Saved schemes	Simplified	Low	Save, note, compare two.
F12 Family dashboard	Simplified	Medium	Member profiles and per-member results; skip co-manager invites.
F13 Notifications	Simplified	Medium	In-app only, triggered by a manual or daily job.
F17 Feedback	Simplified	Low	Thumbs and report form.
F14 Offline-first mode	Future	High	Document in the report; if time allows, demo 10-15 schemes cached.
Future channels and integrations	Future	High	WhatsApp bot, DigiLocker, helper mode, more languages.

About DigiLocker: DigiLocker integration goes through API Setu partner onboarding, which includes identity and purpose checks and can take real calendar time. Some third-party sandboxes also need real user data. For the capstone, keep your own OCR pipeline with sample documents and list DigiLocker as a future data source. If DigiLocker data is added later, it can reduce OCR errors, and OCR would remain the fallback.

8. Useful Resources

Official and documentation links in 8.1 to 8.6 were opened or found in search results on 17 October 2026. Web pages change, so confirm details on the official site before relying on them.

8.1 Government and policy

Resource	Link	How it helps Sarthi
myScheme (official portal)	www.myscheme.gov.in	The main reference product. Study its flow: enter details, search, select and apply; Categories, States and Central Ministries tabs; sign in; language selector.
myScheme FAQs	www.myscheme.gov.in/faqs	Understand what information the official portal shows for each scheme.
API Setu - DigiLocker partner resource centre	apisetu.gov.in/digilocker	Requester API specifications, terms of use and onboarding procedure for future DigiLocker integration.
DigiLocker Requester API Specification v1.12 (PDF)	cf-media.api-setu.in/resources/Requester-APISpecification-V1_12.pdf	Technical details of the consent and document-fetch flow.
DigiLocker Partner Onboarding SOP (PDF)	cf-media.api-setu.in/resources/Partners-SOP.pdf	Explains the steps and checks to become a registered partner.
Setu DigiLocker integration guide	docs.setu.co/data/digilocker/quickstart	Example of a third-party wrapper for DigiLocker flows; useful to understand the request, status and fetch calls.
MeitY - Data Protection Framework (DPDP Act)	www.meity.gov.in/data-protection-framework	Official page for India's data protection law. The DPDP Rules were notified on 13 November 2025; check this page for current commencement details.

8.2 Patent and intellectual property

Resource	Link	How it helps Sarthi
IP India - home	ipindia.gov.in	Official site of the Patent Office with all patent, trademark and copyright services.
Basics of Patents	ipindia.gov.in/basics-of-patents	Plain introduction to what can and cannot be patented in India.
Patent filing process	ipindia.gov.in/page-content/filing-process-1	Step-by-step filing process including provisional and complete specifications.
Public patent search	ipindia.gov.in/patents-before-you-apply-public-search	Entry point to the official search (InPASS) for prior-art checks.
Forms and official fees	ipindia.gov.in/patents-before-you-apply-forms-official-fees	Current form numbers and fees; confirm the reduced fee categories here.
Patent manuals	ipindia.gov.in/resource/patents-resources-manuals	Practice and procedure manuals, useful for the Section 3(k) discussion with the agent.
Electronic Register of Patent Agents	iprsearch.ipindia.gov.in/AgentRegister	Find a registered patent agent.
Startups IP Protection (SIPP) scheme	ipindia.gov.in/page-content/startups-intellectual-property-protection-sipp	Facilitator support for eligible startups; check whether it applies to you.

8.3 Backend, API and database

Resource	Link	How it helps Sarthi
FastAPI documentation	fastapi.tiangolo.com	Main framework for the backend APIs.
FastAPI - OAuth2 with JWT tokens	fastapi.tiangolo.com/tutorial/security/oauth2-jwt/	Token-based sessions for login (F1).
FastAPI - Request Files	fastapi.tiangolo.com/tutorial/request-files/	Handling document uploads (F8).
FastAPI - Background Tasks	fastapi.tiangolo.com/tutorial/background-tasks/	Running OCR and notification jobs without blocking requests.

Resource	Link	How it helps Sarthi
FastAPI - SQL Databases	fastapi.tiangolo.com/tutorial/sql-databases/	Connecting to PostgreSQL with models.
FastAPI - Bigger Applications	fastapi.tiangolo.com/tutorial/bigger-applications/	Organising routers for auth, schemes, eligibility, documents.
FastAPI - Testing	fastapi.tiangolo.com/tutorial/testing/	Writing API tests for the eligibility engine.
FastAPI in Docker	fastapi.tiangolo.com/deployment/docker/	Containerising the backend for deployment.

8.4 OCR and document processing

Resource	Link	How it helps Sarthi
Tesseract OCR repository	github.com/tesseract-ocr/tesseract	Open-source OCR engine with support for more than 100 languages, including Hindi.
Tesseract documentation	tesseract-ocr.github.io/tessdoc/	Installation, command line and API usage.
Improving OCR quality	tesseract-ocr.github.io/tessdoc/ImproveQuality.html	Image pre-processing steps that directly improve accuracy on photographed documents.
Tesseract language data files	tesseract-ocr.github.io/tessdoc/Data-Files.html	Download Hindi and English trained data.

8.5 RAG, vector search and AI assistant

Resource	Link	How it helps Sarthi
FAISS (Facebook AI Similarity Search)	github.com/facebookresearch/faiss	Vector similarity search library for the retrieval step.
FAISS wiki and getting started	github.com/facebookresearch/faiss/wiki	Tutorials and index types.
Chroma documentation	docs.trychroma.com	Alternative open-source vector database with simple Python and JavaScript clients.

8.6 Offline mode and PWA

Resource	Link	How it helps Sarthi
MDN - Service Worker API	developer.mozilla.org/en-US/docs/Web/API/Service_Worker_API	Reference for caching and offline behaviour.
web.dev - PWA training codelab: Going offline	web.dev/codelabs/pwa-training	Hands-on lab to add a service worker and the Cache API to an existing app.

8.7 Similar open-source projects to study before building

These repositories were found in a GitHub search earlier in this project. Read their README and code to understand what is already public, and show them to the patent agent because public projects count as prior art. Repository names and owners are listed as found; please open each link to confirm it is still available.

Resource	Link	How it helps Sarthi
Scheme_Sarathi (jainsanskar0803)	github.com/jainsanskar0803/Scheme_Sarathi	Scheme discovery and eligibility matching; reported to use OCR for documents, voice, and an assisted mode for field workers.
gov-scheme-info (manjukumari98866-collab)	github.com/manjukumari98866-collab/gov-scheme-info	Points-based partial eligibility scoring with deductions; compare with the F8 scoring design.
GovAssist-AI (Kaviya-53)	github.com/Kaviya-53/GovAssist-AI	Explainable rule engine with document OCR.
SchemeSaathi-AI (shauryasinghal)	github.com/shauryasinghal/SchemeSaathi-AI	RAG with FAISS and rule-based eligibility.
Unified-Civic-Scheme-Eligibility-Portal (mahekrathore0728)	github.com/mahekrathore0728/Unified-Civic-Scheme-Eligibility-Portal	Roadmap includes document OCR and identity verification.

Also useful, from their own official documentation sites: PostgreSQL, Next.js, Tailwind CSS, SQLAlchemy and Pytest.

9. Open Items and Disclaimer

Open item	What to do
Project name	The name Sarthi is already used by other government and private products in India (for example a Maharashtra government institute and a UP government app). It is fine for a capstone. Choose a different name before any public launch or trademark filing.
Patent novelty	Similar public projects already show OCR, partial scoring and explainable rules. Keep the claim focused on the specific combination in F8, especially the localized resolution path, and get a professional prior-art search and patent agent review.
Public disclosure	Discuss demo videos, public repositories and presentations with the patent agent before publishing the F8 module.
Local data	Collect and verify office, process and time data for at least one state and a few districts from official sources before demonstrating F7 and F8.
Data protection	Get a legal review of consent text, children's data, document retention and the Aadhaar masking approach before a real launch.
DigiLocker	Plan partner onboarding early if it will be used after the capstone; it needs approval time.

Disclaimer. *This is a technical and product-planning document for a capstone project. It is not legal advice, a patentability opinion or a filing-ready patent specification. Sarthi gives preliminary guidance only; the concerned government authority makes the final eligibility decision.*